

Executive Project Summary Report

End-to-End E-Commerce Data Extraction, EDA & Analytics Dashboard

AUTHOR

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TARGET SITE

Books to Scrape

DATASET SIZE

100 Records / 5 Fields

PIPELINE STATUS

100% Production Ready

Executive Note: This document summarizes the complete data pipeline designed to automate product data collection, inspect categorical structure, and deliver leadership dashboard analytics to identify pricing optimizations.

1. Executive Summary

This project demonstrates a complete, end-to-end data pipeline lifecycle designed to extract, clean, analyze, and visualize e-commerce inventory data. Using custom Python web scraping techniques, an unstructured web dataset was transformed into a structured 100-record CSV dataset containing pricing, availability, and customer rating attributes.

Through Exploratory Data Analysis (EDA) and Data Visualization, key operational insights were uncovered regarding catalog price distribution, inventory balancing, and product rating metrics. The findings offer immediate strategic pricing opportunities for inventory optimization.

2. Key Technical Deliverables

Task 1: Web Scraping Pipeline Architecture

The extraction component utilizes Python's `requests` and `BeautifulSoup` to crawl multi-page catalogs safely and reliably.

Scrape Speed

1.0s / page

Throttled via `time.sleep()`

Character Encoding

UTF-8

Regex price sanitization

Data Integrity

0% Nulls

0 Duplicate records

Pipeline Stage	Technical Solution	Business Impact
HTTP Requests	<code>requests</code> with custom User-Agent headers	Bypasses basic anti-bot blocking filters.
Parsing & Regex	<code>BeautifulSoup + re.sub(r"[^\d.]", "", price)</code>	Sanitizes misencoded characters (e.g., <code>Â£</code>) into floats.
Pagination & Export	Looping page templates to <code>pandas.DataFrame.to_csv()</code>	Delivers a clean CSV dataset for analytics ingestion.

3. Analytics & Executive Dashboard

Below is a rendered representation of the 4-panel executive dashboard built with Matplotlib and Seaborn, designed to evaluate inventory pricing across customer star ratings.



4. Key Analytical Insights & Recommendations

- **Uncoupled Pricing Strategy:** Book pricing operates independently of customer ratings. Higher-rated items (4-star and 5-star) are not priced at a premium compared to lower-rated titles.
- **Margin Optimization Potential:** 1-Star books exhibit the highest mean price (~£40.09), while 5-star books average £35.10. Introducing value-based dynamic pricing could significantly boost overall margin.
- **Inventory Turnover Strategy:** High-priced, low-rated books (>£40.00 with 1 star) should be targeted for promotional clearance discounts to reduce holding costs.

5. Project File Repository Architecture

```
├─ scrape_books.py # Task 1: Web scraping Python script ─ books_scraped_dataset.csv #
Task 1: Cleaned scraped CSV dataset ─ eda_analysis.py # Task 2: Exploratory data analysis
script ─ generate_visuals.py # Task 3: Dashboard generation script ─
executive_dashboard.png # Task 3: Saved high-resolution dashboard ─
executive_summary_report.html # Task 4: Print-ready PDF report
```